MATRIX AND LINEAR ALGEBRA

PYQ (2022) → [Marks → 20]

Q.1(c) If $\begin{bmatrix} 4/3 \\ 1 \end{bmatrix}$ is an eigenvector of $\begin{bmatrix} 2 & 4 \\ 3 & 1 \end{bmatrix}$.
what is the associated eigenvalue?

(i) 4/3 (ii) 5 (iii) −2 (iv) None of the above

→ Given;

let $A = \begin{bmatrix} 2 & 4 \\ 3 & 1 \end{bmatrix}$

The characteristic equation of matrix is

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 2-\lambda & 4 \\ 3 & 1-\lambda \end{vmatrix} = 0$$

$$(2-\lambda)(1-\lambda) - 12 = 0$$
$$2 - \lambda - 2\lambda + \lambda^2 - 12 = 0$$
$$\lambda^2 - 3\lambda - 10 = 0$$
$$\lambda^2 - 5\lambda + 2\lambda - 10 = 0$$
$$\lambda(\lambda - 5) + 2(\lambda - 5) = 0$$
$$(\lambda - 5)(\lambda + 2) = 0$$
$$\lambda = 5 \text{ or } \lambda = -2$$

at $\lambda = 5$

$$[A - \lambda I][X] = 0$$

$$\begin{bmatrix} 2-\lambda & 4 \\ 3 & 1-\lambda \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$\frac{y}{3/4}$ $\frac{y}{3/3}$

Q.1

$$\begin{bmatrix} -3 & 4 \\ 3 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$-3x+4y = 0$

$3x-4y = 0$

$3x = 4y$

$$\frac{x}{1} = \frac{4y}{3}$$

$$\frac{x}{1} = \frac{y}{3/4}$$

$$\frac{3x}{4} = \frac{y}{1}$$

$$\frac{x}{4/3} = \frac{y}{1}$$

$\therefore$ Eigen vector at $\underline{\lambda = 5}$ is $\begin{bmatrix} 4/3 \\ 1 \end{bmatrix}$

$\therefore$ associated eigen value is 5

Q.1.(b) If $3x+2y+z=0$, $x+4y+z=0$, $2x+y+4z=0$ be a system of equation, then

(i) It is inconsistent

(ii) It has only the trivial solution ($x=y=z=0$)

(iii) Determinant of the matrix of coefficient is zero.

(iv) None of these.

⇒ Given;

$$3x+2y+z=0$$
$$x+4y+z=0$$
$$2x+y+4z=0$$

let $A = \begin{bmatrix} 3 & 2 & 1 \\ 1 & 4 & 1 \\ 2 & 1 & 4 \end{bmatrix}$ be a coefficient matrix.

∴ • If $|A|=0$ then system of equation ~~is~~ has infinitly many soln.

• If $|A| \neq 0$ then system of equation has trivial soln.

$$|A| = \begin{vmatrix} 3 & 2 & 1 \\ 1 & 4 & 1 \\ 2 & 1 & 4 \end{vmatrix}$$

$$= 3[16-1] - 2[4-2] + 1[1-8]$$

$$= 45 - 4 - 7$$

$$|A| = 34$$

$|A| \neq 0$ ∴ It has only trivial soln ($x=y=z=0$)

Q.) 1.(g).) The rank of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$ is

(a) 3 (b) 2 ✓ (c) 1 (d) None of these

⇒ Given;

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$$

apply $R_2 \to R_2 - 2R_1$
$R_3 \to R_3 - 3R_1$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

∴ Rank of matrix is 1

Q.7)(a) Find the values of a, b, c if $A = \begin{bmatrix} 0 & 2b & c \\ a & b & -c \\ a & -b & c \end{bmatrix}$ is orthogonal?

$\Rightarrow$ A matrix is orthogonal if

$$A^{-1} = A^T$$

we know,

$$A \cdot A^{-1} = I$$

$$A \cdot A^T = I \qquad [A^{-1} = A^T]$$

$$A = \begin{bmatrix} 0 & 2b & c \\ a & b & -c \\ a & -b & c \end{bmatrix}$$

$$A^T = \begin{bmatrix} 0 & a & a \\ 2b & b & -b \\ c & -c & c \end{bmatrix}$$

$$A \cdot A^T = \begin{bmatrix} 0 & 2b & c \\ a & b & -c \\ a & -b & c \end{bmatrix} \begin{bmatrix} 0 & a & a \\ 2b & b & -b \\ c & -c & c \end{bmatrix}$$

$$A \cdot A^T = \begin{bmatrix} 4b^2+c^2 & 2b^2-c^2 & c^2-2b^2 \\ 2b^2-c^2 & a^2+b^2+c^2 & a^2-b^2-c^2 \\ c^2-2b^2 & a^2-b^2-c^2 & a^2+b^2+c^2 \end{bmatrix}$$

$A \cdot A^T = I$

$$\begin{bmatrix} 4b^2+c^2 & 2b^2-c^2 & c^2-2b^2 \\ 2b^2-c^2 & a^2+b^2+c^2 & a^2-b^2-c^2 \\ c^2-2b^2 & a^2-b^2-c^2 & a^2+b^2+c^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$4b^2+c^2 = 1$ —— (i)
$2b^2-c^2 = 0$ —— (ii)
$c^2-2b^2 = 0$ —— (iii)
$a^2+b^2+c^2 = 1$ —— (iv)
$a^2-b^2-c^2 = 0$ —— (v)

from eq- (ii) and (iii)

$c^2 = 2b^2$ —— (vi)

from eq- (v)

$a^2 = b^2+c^2$ —— (vii)

put the value $c^2 = 2b^2$ in eq-(vii)

$a^2 = b^2+2b^2$
$a^2 = 3b^2$ —— (viii)

put the value of eq-(vi) and (viii) in eq-(iv)

$$3b^2 + b^2 + 2b^2 = 1$$

$$5b^2 = 1$$

$$b^2 = \frac{1}{5}$$

$$b = \pm \frac{1}{\sqrt{5}}$$

$$a^2 = 3b^2$$

$$a^2 = \frac{3}{5}$$

$$a = \pm\sqrt{\frac{3}{5}}$$

$$c = \pm\sqrt{\frac{2}{5}}$$

[Hence, $a = \pm\sqrt{\frac{3}{5}}$, $b = \frac{\pm 1}{\sqrt{5}}$, $c = \pm\sqrt{\frac{2}{5}}$]

Q.7(b) Verify Cayley-Hamilton theorem for the matrix

$$A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$$ and find its inverse. Also express

$$A^5 - 4A^4 - 7A^3 + 11A^2 - A - 10I$$ as a linear polynomial in A.

⇒ Given;

$$A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$$

The characteristic equation of matrix is

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 1-\lambda & 4 \\ 2 & 3-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)(3-\lambda) - 8 = 0$$

$$3 - 3\lambda - \lambda + \lambda^2 - 8 = 0$$

$$\lambda^2 - 4\lambda - 5 = 0$$

So verify cayley hamilton theorem; the equation we get after solving characteristic equation

$$\lambda^2 - 4\lambda - 5 = 0$$

its verify if and only if $\lambda = A$ satisfy the equation

$$A^2 - 4A - 5I = 0$$

$$A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 9 & 16 \\ 8 & 17 \end{bmatrix}$$

$$4A = \begin{bmatrix} 4 & 16 \\ 8 & 12 \end{bmatrix}$$

$$5I = \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}$$

$$A^2 - 4A - 5I$$

$$\begin{bmatrix} 9 & 16 \\ 8 & 17 \end{bmatrix} - \begin{bmatrix} 4 & 16 \\ 8 & 12 \end{bmatrix} - \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Hence its verify cayley hamilton theorem.

Now, to express

$A^5 - 4A^4 - 7A^3 + 11A^2 - A - 10I$ as linear polynomial in A.

$$
\begin{array}{r|l}
 & A^3 - 2A + 3 \\
\hline
A^2 - 4A - 5I & A^5 - 4A^4 - 7A^3 + 11A^2 - A - 10I \\
 & -A^5 + 4A^4 + 5A^3 \\
\hline
 & -2A^3 + 11A^2 - A \\
 & -2A^3 + 8A^2 + 10A \\
\hline
 & 3A^2 - 11A - 10I \\
 & 3A^2 - 12A - 15I \\
\hline
 & A + 5I
\end{array}
$$

$\therefore$ we can write $A^5 - 4A^4 - 7A^3 + 11A^2 - A - 10I$ in form of

$$A^5 - 4A^4 - 7A^3 + 11A^2 - A - 10I = (A^2 - 4A - 5I) \times (A^3 - 2A + 3) + A + 5I$$

$$A^2 - 4A^4 - 7A^3 + 11A^2 - A - 10I = A + 5I$$

$$(A^2 - 4A - 5I = 0)$$

$\therefore$ we can [redacted] write in linear polynomial;

$$\underline{A + 5I}$$

Expected Ques (2023-24)

Q.1) If $\det(A) = 7$, where

$$A = \begin{bmatrix} a & b & c \\ e & e & f \\ g & w & e \end{bmatrix}$$ then $\det(2A)^{-1}$ is equal to

(i) $1/14$ (ii) $1/49$ ✓(iii) $1/56$ (iv) $7/2$

→ Given;

$|A| = 7$

$|2A^{-1}| = ?$

We know; $|KA^{-1}| = \dfrac{1}{|KA|} = \dfrac{1}{K^n|A|}$

[n = order]

[k = constant]

$$|2A^{-1}| = \frac{1}{|2A|} = \frac{1}{2^3|A|} = \frac{1}{8 \times 7} = \frac{1}{56}$$

Q.2. let

$$N = \begin{bmatrix} 3/5 & -4/5 & 0 \\ 4/5 & 3/5 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$, then N is

(i) non-invertible (ii) Skew-Symmetric

(iii) Symmetric ✓(iv) orthogonal

⇒

$A = A^T \rightarrow$ Symmetric

$$A^T = \begin{bmatrix} 3/5 & 4/5 & 0 \\ -4/5 & 3/5 & 0 \\ 0 & 0 & 1 \end{bmatrix} \neq A$$

$A = -A^T \longrightarrow$ Skew-symmetric

$$-A^T = \begin{bmatrix} -3/5 & -4/5 & 0 \\ 4/5 & -3/5 & 0 \\ 0 & 0 & -1 \end{bmatrix} \neq A$$

If $|A| \neq 0$ then $\rightarrow$ [redacted] invertible $\left[A^{-1} = \frac{adj(A)}{|A|}\right]$

If $|A| = 0$ then $\rightarrow$ Non-invertible

$$|A| = \begin{vmatrix} 3/5 & -4/5 & 0 \\ 4/5 & 3/5 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

$$= 3/5\left[3/5 - 0\right] + \frac{4}{5}\left[4/5 - 0\right] + 0\left[0 - 0\right]$$

$$= \frac{9}{25} + \frac{16}{25} = \frac{25}{25} = 1$$

$|A| \neq$ $\therefore$ It can invertible.

$\therefore$ last option orthogonal;

$$A^{-1} = A^T$$

$$A^{-1} \begin{bmatrix} 3/5 & 4/5 & 0 \\ -4/5 & 3/5 & 0 \\ 0 & 0 & 1 \end{bmatrix} = A^T$$

Q.2. The system of linear equation $x - y + 3z = 4$, $x + z = 2$, $x + y - z = 0$ has

(i) Infinetly many solutions
(ii) finitly many solution
(iii) a unique solution
(iv) no solution

⇒

Given;

$$x - y + 3z = 4$$
$$x + z = 2$$
$$x + y - z = 0$$

let $A = \left[\begin{array}{ccc:c} 1 & -1 & 3 & 4 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -1 & 0 \end{array}\right]$ be augment matrix

Apply $R_2 \to R_2 - R_1$
$R_3 \to R_3 - R_1$

$$A = \left[\begin{array}{ccc:c} 1 & -1 & 3 & 4 \\ 0 & 1 & -2 & -2 \\ 0 & 2 & -4 & -4 \end{array}\right]$$

Apply $R_3 \to R_3 - 2R_2$

$$A = \left[\begin{array}{ccc:c} 1 & -1 & 3 & 4 \\ 0 & 1 & -2 & -2 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

Hence Rank of matrix = Rank of augmented matrix < no. of unknown

∴ Infinitly many sol^n

Q.4. If A and B are any ~~two~~ ~~tw~~ two square matrices of order 2×2, then $(A+B)^2$ is equal to

(i) $A^2 + 2AB + B^2$

(ii) $A^2 + AB - BA + B^2$

~~(iii)~~ $A^2 + AB + BA + B^2$

(iv) $A^2 + 2BA + B^2$

$\Rightarrow$

$$(A+B)^2 = (A+B)(A+B)$$

$$\underline{A^2 + BA + AB + B^2}$$

[because $AB \neq \underline{BA}$]

Q.5. Find the value of k for which the system of equation $(3k-8)x + 3y + 3z = 0$, $3x + (3k-8)y + 3z = 0$, $3x + 3y + (3k-8)z = 0$ has a non-trivial solution.

$\Rightarrow$ Given,

$$(3k-8)x + 3y + 3z = 0$$
$$3x + (3k-8)y + 3z = 0$$
$$3x + 3y + (3k-8)z = 0$$

for non-trivial solution then it should be infinetly many soln.

for infinetly many solution

determinant of coefficient matrix is $= 0$

$33\,6$ $\quad 5\times11\times5\times3$ $\quad 2\,2\,1\,5$

$$\begin{vmatrix} 3k-8 & 3 & 3 \\ 3 & 3k-8 & 3 \\ 3 & 3 & 3k-8 \end{vmatrix} = 0$$

$$(3k-8)\left[(3k-8)(3k-8)-9\right] - 3\left[3(3k-8)-9\right] + 3\left[9-3(3k-8)\right] = 0$$

$$(3k-8)\left[9k^2+64-48k-9\right] - 3\left[9k-24-9\right] + 3\left[9-9k+24\right] = 0$$

$$(3k-8)\left[9k^2-48k+55\right] - 3\left[9k-33\right] + 3\left[33-9k\right] = 0$$

$$(3k-8)\left[9k^2-33k-15k+55\right] + 3(33-9k) + 3(33-9k) = 0$$

$$(3k-8)\left[3k(3k-11)-5k(3k-11)\right] + 6(33-9k) = 0$$

$$(3k-8)\left[(3k-11)(3k-5)\right] + 18(11-3k) = 0$$

$$(3k-11)\left[(3k-8)(3k-5)-18\right] = 0$$

$$(3k-11)\left[9k^2-24k-15k+40-18\right] = 0$$

$$(3k-11)\left[9k^2-39k+22\right] = 0$$

$$(3k-11)\left[9k^2-33k-6k+22\right] = 0$$

$$(3k-11)\left[3k(3k-11)-2(3k-11)\right] = 0$$

$$(3k-11)(3k-11)(3k-2) = 0$$

$$k = 11/3,\ 11/3,\ 2/3$$

~~Q.6.) If X is the eigenvector of matrix A, then which is not the eigenvector of A^{-1}?~~

~~(i) X (ii) X^{-1} (iii) X (iv) 2X~~

~~⇒~~

Q.6.) If λ is an eigenvalue of matrix A then $\frac{1}{\lambda}$ is the eigen value of

(i) A^2 (ii) A^{-1} (iii) A^2 (iv) A

⇒

$A \rightarrow$ matrix $\rightarrow$ Eigenvalue $\rightarrow \lambda$

$A^{-1} \rightarrow$ matrix $\rightarrow$ Eigenvalue $\rightarrow \frac{1}{\lambda}$

Q.7.) If the $\lambda_1, \lambda_2, \lambda_3, \cdots \lambda_n$ are the eigen value of the matrix A, then $\lambda_1 - 2, \lambda_2 - 2, \lambda_3 - 2, \lambda_4 - 2 \cdots \lambda_n - 2$ shall be the eigen value of the matrix

(i) A (ii) A − I (iii) A − 2I (iv) A − 3I

⇒

$\lambda_1, \lambda_2, \lambda_3 \cdots \lambda_n$ are eigen value of matrix A

$\lambda_1 - 2, \lambda_2 - 2, \lambda_3 - 2, \lambda_4 - 2, \cdots \lambda_n - 2$ are the eigen value of matrix

A − 2I

formula; A − mI for → $\lambda_1 - m, \lambda_2 - m, \cdots \lambda_n - m$

Q8) The eigen value of an idempotent matrix is/ are?

(i) 0 (ii) 1 (iii) both (iv) none of the above

=> Idempotent matrix

$$\boxed{A = A^2}$$

∴ It can 0 or 1.

Q.9) Determine the value of P such that rank of matrix

$$\begin{bmatrix} 1 & 1 & -1 & 0 \\ 4 & 4 & -3 & 1 \\ P & 2 & 2 & 2 \\ 9 & 9 & P & 3 \end{bmatrix}$$ is 3.

=>

$$\begin{bmatrix} 1 & 1 & -1 & 0 \\ 4 & 4 & -3 & 1 \\ P & 2 & 2 & 2 \\ 9 & 9 & P & 3 \end{bmatrix}$$

Applying $R_2 \to R_2 - 4R_1$
$R_3 \to R_3 - 2R_1$
$R_4 \to R_4 - 9R_1$

$$\begin{bmatrix} 1 & 1 & -1 & 0 \\ 0 & 0 & 1 & 1 \\ P-2 & 0 & 4 & 2 \\ 0 & 0 & P+1 & 3 \end{bmatrix}$$

Applying $R_4 \to R_4 - 3R_2$

$$\begin{bmatrix} 1 & 1 & -1 & 0 \\ 0 & 0 & 1 & 1 \\ p-2 & 0 & 4 & 2 \\ 0 & 0 & p+6 & 0 \end{bmatrix}$$

Rank is 3 if;

$$p+6=0$$

$$\boxed{p=-6}$$

Q.10.) Find the characterstic equation of the matrix,

VVI

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix} \text{ and}$$

hence compute A^{-1}. Also find the matrix represented by

$$A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I.$$

$\Rightarrow$ Given;

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$$

The characterstic equation of given matrix is

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 2-\lambda & 1 & 1 \\ 0 & 1-\lambda & 0 \\ 1 & 1 & 2-\lambda \end{vmatrix} = 0$$

$$(2-\lambda)[(1-\lambda)(2-\lambda)-0] - 1[0-0] + 1[0-(1-\lambda)] = 0$$

$$(2-\lambda)[2-2\lambda-\lambda+\lambda^2] + (\lambda-1) = 0$$

$$(2-\lambda)[\lambda^2-3\lambda+2] + \lambda-1 = 0$$

$$2\lambda^2 - \lambda^3 - 6\lambda + 3\lambda^2 + 4 - 2\lambda + \lambda - 1 = 0$$

$$-\lambda^3 - 7\lambda + 5\lambda^2 + 3 = 0$$

$$\lambda^3 - 5\lambda^2 + 7\lambda - 3 = 0$$

To verify cayley-hamilton theorem, at

$$\lambda = A$$

$$A^3 - 5A^2 + 7A - 3I = 0$$

$$A^2 = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 5 & 4 & 4 \\ 0 & 1 & 0 \\ 4 & 4 & 5 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 5 & 4 & 4 \\ 0 & 1 & 0 \\ 4 & 4 & 5 \end{bmatrix} \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 14 & 13 & 13 \\ 0 & 1 & 0 \\ 13 & 13 & 14 \end{bmatrix}$$

$$5A^2 = \begin{bmatrix} 25 & 20 & 20 \\ 0 & 5 & 0 \\ 20 & 20 & 25 \end{bmatrix}$$

$$7A = \begin{bmatrix} 14 & 7 & 7 \\ 0 & 7 & 0 \\ 7 & 7 & 14 \end{bmatrix}$$

$$3I = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

$$A^3 - 5A^2 + 7A - 3I = 0$$

$$\begin{bmatrix} 14 & 13 & 13 \\ 0 & 1 & 0 \\ 13 & 13 & 14 \end{bmatrix} - \begin{bmatrix} 25 & 20 & 20 \\ 0 & 5 & 0 \\ 20 & 20 & 25 \end{bmatrix} + \begin{bmatrix} 14 & 7 & 7 \\ 0 & 7 & 0 \\ 7 & 7 & 14 \end{bmatrix}$$

$$- \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Hence it's verify ~~ut~~ cayley hamilton theorem;

Now to find A^{-1}

Given eq^n

$$A^3 - 5A^2 + 7A - 3I = 0$$

multiply this equation by A^{-1}, we get;

$$A^2 - 5A + 7I - 3A^{-1} = 0$$

$$A^{-1} = \frac{1}{3}\left[A^2 - 5A + 7I\right]$$

$$= \frac{1}{3}\left[\begin{bmatrix} 5 & 4 & 4 \\ 0 & 1 & 0 \\ 4 & 4 & 5 \end{bmatrix} - \begin{bmatrix} 10 & 5 & 5 \\ 0 & 5 & 0 \\ 5 & 5 & 10 \end{bmatrix} + \begin{bmatrix} 7 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 7 \end{bmatrix}\right]$$

$$\cancel{\frac{1}{3}\begin{bmatrix} 2 & -5 & -1 & -1 \\ 0 & -4 & 0 \\ -1 & -1 & -5 \end{bmatrix}}$$

$$A^{-1} = \frac{1}{3}\begin{bmatrix} 2 & -1 & -1 \\ 0 & 3 & 0 \\ -1 & -1 & 2 \end{bmatrix}$$

Now reducing the polynomial;

$$A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I$$

divide this polynomial by

$$A^3 - 5A^2 + 7A - 3I \text{ , we get;}$$

$$\begin{array}{r|l} & A^5 + A \\ \hline A^3 - 5A^2 + 7A - 3I & A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I \\ & \underline{A^8 - 5A^7 + 7A^6 - 3A^5} \\ & - \quad + \quad - \quad + \\ & A^4 - 5A^3 + 8A^2 - 2A + I \\ & \underline{A^4 - 5A^3 + 7A^2 - 3A} \\ & - \quad + \quad - \quad + \\ & A^2 + A + I \end{array}$$

Hence we can represent above polynomial by;

$$A^2 + A + I.$$

$$A^2 + A + I = \begin{bmatrix} 5 & 4 & 4 \\ 0 & 1 & 0 \\ 4 & 4 & 5 \end{bmatrix} + \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A^2 + A + I = \begin{bmatrix} 8 & 5 & 5 \\ 0 & 3 & 0 \\ 5 & 5 & 8 \end{bmatrix} \text{ ///}$$

Q.11. If $A = \frac{1}{3}\begin{bmatrix} 1 & 2 & a \\ 2 & 1 & b \\ 2 & -2 & c \end{bmatrix}$ is orthogonal, find a, b, c and A^{-1}.

$\Rightarrow$ For a matrix be orthogonal;

$$A^{-1} = A^T$$

We know; $A \cdot A^{-1} = I$

$$A \cdot A^T = I \qquad [A^{-1} = A^T]$$

$$A^T = \frac{1}{3}\begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ a & b & c \end{bmatrix}$$

$$A \cdot A^T = \frac{1}{3}\begin{bmatrix} 1 & 2 & a \\ 2 & 1 & b \\ 2 & -2 & c \end{bmatrix} \frac{1}{3}\begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ a & b & c \end{bmatrix}$$

$$= \frac{1}{9}\begin{bmatrix} a^2+5 & 4+ab & ac-2 \\ 4+ab & 5+b^2 & 2+bc \\ ac-2 & 2+bc & 8+c^2 \end{bmatrix}$$

$$A \cdot A^T = I$$

$$\frac{1}{9}\begin{bmatrix} a^2+5 & 4+ab & ac-2 \\ 4+ab & 5+b^2 & 2+bc \\ ac-2 & 2+bc & 8+c^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} a^2+5 & 4+ab & ac-2 \\ 4+ab & b^2+5 & 2+bc \\ ac-2 & 2+bc & 8+c^2 \end{bmatrix} = \begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix}$$

$a^2+5=9$ —— (i)
$ab+4=0$ —— (ii)
$ac-2=0$ —— (iii)
$b^2+5=9$ —— (iv)
$2+bc=0$ —— (v)
$8+c^2=9$ —— (vi)

from eq- (i)

$a^2+5=9$
$a^2=4$
$a=\pm 2$

from eq — (iv)

$b^2+5=9$
$b^2=4$
$b=\pm 2$

from eq- (vi)

$8+c^2=9$
$c^2=1$
$c=\pm 1$

Hence value of ;

$$a = \pm 2$$
$$b = \pm 2$$
$$c = \pm 1$$

$A^{-1} = ?$

We know, $A^T = A^{-1}$ [for orthogonal]

$$A^T = \frac{1}{3}\begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ a & b & c \end{bmatrix}$$

$$A^T = \frac{1}{3}\begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ \pm 2 & \pm 2 & \pm 1 \end{bmatrix}$$

Hence,

$$A^{-1} = \frac{1}{3}\begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ \pm 2 & \pm 2 & \pm 1 \end{bmatrix}$$

Q.12. If $A = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$, find A^{-1}. Also find two non-singular matrix P and Q such that $PAQ = I$, where I is a unit matrix and verify that $A^{-1} = QP$.

Given;

$$A = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$$

$$A^{-1} = \frac{adj(A)}{|A|}$$

Now,

$$|A| = \begin{vmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{vmatrix}$$

$$= 3[-3+4] + 3[2-0] + 4[-2+0]$$

$$= 3+6-8$$

$$|A| = 1$$

Now, Minor (M_{ij})

$M_{11} = 1$

$M_{12} = 2$

$M_{13} = -2$

$M_{21} = 1$

$M_{22} = 3$

$M_{23} = -3$

$M_{31} = 0$

$M_{32} = 4$

$M_{33} = -3$

$$M = \begin{bmatrix} 1 & 2 & -2 \\ 1 & 3 & -3 \\ 0 & 4 & -3 \end{bmatrix}$$

cofactor of $A = \begin{bmatrix} 1 & -2 & -2 \\ -1 & 3 & 3 \\ 0 & -4 & -3 \end{bmatrix}$

adj(A) = Transpose of cofactor

$$\text{adj}(A) = \begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & -4 \\ -2 & 3 & -3 \end{bmatrix}.$$

$$A^{-1} = \frac{\text{adj}(A)}{|A|}$$

$$A^{-1} = \frac{\begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & -4 \\ -2 & 3 & -3 \end{bmatrix}}{1}$$

Hence, $A^{-1} = \begin{bmatrix} +1 & -1 & 0 \\ -2 & 3 & -4 \\ -2 & 3 & -3 \end{bmatrix}$

We know,

$$A = AI$$
$$A = IAI$$

$$\begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

To find two matrix P and Q such
$PAQ = I$
then we transform A into I
we operated only row operation into P
and operates only column operation int Q.

Now apply $R_1 \to R_1 - R_2$

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$R_2 \to R_2 - 2R_1$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Apply $C_2 \to C_2 + C_3$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 4 \\ 0 & +0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & +1 & 1 \end{bmatrix}$$

Apply $C_3 \to C_3 - 4C_2$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -4 \\ 0 & 1 & -3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -4 \\ 0 & 1 & -3 \end{bmatrix}$$

Here;

$$I = PAQ \text{ form;}$$

where

$$P = \begin{bmatrix} +1 & -1 & 0 \\ -2 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$Q = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -4 \\ 0 & 1 & -3 \end{bmatrix}$$

Now to verify $A^{-1} = QP$

$$Q \cdot P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -4 \\ 0 & 1 & -3 \end{bmatrix} \times \begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$Q \cdot P = \begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & -4 \\ -2 & 3 & -3 \end{bmatrix} = A^{-1}$$

Hence verify, $A^{-1} = Q \cdot P$

Q.13. Verify cayley-hamilton theorem for the matrix

V.V.I

$$\begin{bmatrix} 0 & 0 & 1 \\ 3 & 1 & 0 \\ -2 & 1 & 4 \end{bmatrix}$$

$\Rightarrow$ Criven;

let $A = \begin{bmatrix} 0 & 0 & 1 \\ 3 & 1 & 0 \\ -2 & 1 & 4 \end{bmatrix}$

~~To verify cayley thm~~

The characterstic equation of given matrix is

$|A - \lambda I| = 0$

$$\begin{vmatrix} 0-\lambda & 0 & 1 \\ 3 & 1-\lambda & 0 \\ -2 & 1 & 4-\lambda \end{vmatrix} = 0$$

$(-\lambda)\left[(4-\lambda)(1-\lambda) - 0\right] - 0 + 1\left[3 + 2(1-\lambda)\right] = 0$

$(-\lambda)\left[4 - \lambda - 4\lambda + \lambda^2\right] + \left[3 + 2 - 2\lambda\right] = 0$

$(-\lambda)\left[\lambda^2 - 5\lambda + 4\right] + 5 - 2\lambda = 0$

$-\lambda^3 + 5\lambda^2 - 4\lambda + 5 - 2\lambda = 0$

$-\lambda^3 + 5\lambda^2 - 6\lambda + 5 = 0$

$\lambda^3 - 5\lambda^2 + 6\lambda - 5 = 0$

To verify cayley hamilton theorem;

the given equation also verifys why $\lambda = A$

$\therefore \quad A^3 - 5A^2 + 6A - 5I = 0$

Now,

$$A^2 = \begin{bmatrix} 0 & 0 & 1 \\ 3 & 1 & 0 \\ -2 & 1 & 4 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 3 & 1 & 0 \\ -2 & 1 & 4 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} -2 & 1 & 4 \\ 3 & 1 & 3 \\ -5 & 5 & 14 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} -2 & 1 & 4 \\ 3 & 1 & 3 \\ -5 & 5 & 14 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 3 & 1 & 0 \\ -2 & 1 & 4 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} -5 & +5 & 14 \\ -3 & 4 & 15 \\ -13 & 19 & 51 \end{bmatrix}$$

$$5A^2 = \begin{bmatrix} -10 & 5 & 20 \\ 15 & 5 & 15 \\ -25 & 25 & 70 \end{bmatrix}$$

$$6A = \begin{bmatrix} 0 & 0 & 6 \\ 18 & 6 & 0 \\ -12 & 6 & 24 \end{bmatrix}$$

$$5I = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

$$A^3 - 5A^2 + 6A - 5I = 0$$

$$\begin{bmatrix} -5 & +5 & 14 \\ -3 & 4 & 15 \\ -13 & 19 & 51 \end{bmatrix} - \begin{bmatrix} -10 & 5 & 20 \\ 15 & 5 & 15 \\ -25 & 25 & 70 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 6 \\ 18 & 6 & 0 \\ -12 & 6 & 24 \end{bmatrix}$$

$$- \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Hence verified cayley hamilton theorem.

Q.14 Investigate for what values of a and b, the equation $x+y+z=6$, $x+2y+3z=10$, $x+2y+az=b$ have (i) no solution (ii) a unique solution (iii) an infinite number of solution.

VVI

$\Rightarrow$ Given;

$$x+y+z=6$$
$$x+2y+3z=10$$
$$x+2y+az=b$$

let $K = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 2 & a \end{bmatrix}$ be coefficient matrix.

$$A = \begin{bmatrix} 1 & 1 & 3 & : & 6 \\ 1 & 2 & 3 & : & 10 \\ 1 & 2 & a & : & b \end{bmatrix}$$ be

augmented matrix

for augmented matrix;

Apply
$R_2 \to R_2 - R_1$
$R_3 \to R_3 - R_1$

$$A = \begin{bmatrix} 1 & 1 & 3 & : & 6 \\ 0 & +1 & 0 & : & 4 \\ 0 & 1 & a-3 & : & b-6 \end{bmatrix}$$

apply $R_3 \to R_3 - R_2$

$$A = \begin{bmatrix} 1 & 1 & 3 & : & 6 \\ 0 & 1 & 0 & : & 4 \\ 0 & 0 & a-3 & : & b-10 \end{bmatrix}$$

for

(i) No solution;

$\rho(K) \neq \rho(A)$
(rank of coefficent matrix $\neq$ rank of augmented matrix)

Then it possible when,

$\boxed{\begin{matrix} a=3 \\ b \neq 10 \end{matrix}}$ but

(ii) for unique solution

$$\rho(A) = \rho(K) = n(3)$$

(rank of augmented matrix = rank of coefficient matrix = no. of unknowns.

then it possible when,

$$\boxed{\begin{matrix} a \neq 3 \\ b \neq 10 \end{matrix}} \text{ and}$$

(iii) for infinetly many solution;

$$\rho(K) = \rho(A) < n(3)$$

(rank of coefficient matrix = rank of augmented matrix < no. of unknown)

then it possible when

$$\boxed{\begin{matrix} a = 3 \\ b = 10 \end{matrix}} \text{ and}$$

Q.15. Show that the equations $3x+4y+5z=a$, $4x+5y+6z=b$ and $5x+6y+7z=c$ do not have a solution unless $a+c=2b$.

$\Rightarrow$ Given;

$$3x+4y+5z=a$$
$$4x+5y+6z=b$$
$$5x+6y+7z=c$$

let $K = \begin{bmatrix} 3 & 4 & 5 \\ 4 & 5 & 6 \\ 5 & 6 & 7 \end{bmatrix}$ be coefficient matrix

$A = \left[\begin{array}{ccc:c} 3 & 4 & 5 & a \\ 4 & 5 & 6 & b \\ 5 & 6 & 7 & c \end{array}\right]$ be

augmented matrix

for augment matrix (A)

apply $R_2 \to 3R_2 - 4R_1$
$R_3 \to 3R_3 - 5R_1$

$$A = \left[\begin{array}{ccc:c} 3 & 4 & 5 & a \\ 0 & -1 & -2 & 3b-4a \\ 0 & -2 & -4 & 3c-5a \end{array}\right]$$

Apply $R_3 \to R_3 - 2R_2$

$$A = \left[\begin{array}{ccc:c} 3 & 4 & 5 & a \\ 0 & -1 & -2 & 3b-4a \\ 0 & 0 & 0 & (3c-5a)-2(3b-4a) \end{array}\right]$$

if $(3c-5a) - 2(3b-4a) = 0$, then only solution possible

after solving we get

$3c - 5a - 6b + 8a = 0$

$3a+3c-6b=0$
$a+c-2b=0$
$2b=a+c$

Hence it's verified that

unless $(a+c=2b)$, ~~[illegible]~~
the equation has no solution.

Q.16. For what value of λ and μ do the system of equation $x+y+z=6$, $x+2y+3z=10$ and $x+2y+\lambda z=\mu$ have
(i) no solution, (ii) unique solution and (iii) more than one solution.

$\Rightarrow$ Given;

$$x+y+z=6$$
$$x+2y+3z=10$$
$$x+2y+\lambda z=\mu$$

let
$$R=\begin{bmatrix}1 & 1 & 1\\ 1 & 2 & 3\\ 1 & 2 & \lambda\end{bmatrix}$$ be coefficient matrix

$$A=\left[\begin{array}{ccc:c}1 & 1 & 1 & 6\\ 1 & 2 & 3 & 10\\ 1 & 2 & \lambda & \mu\end{array}\right]$$ be augmented matrix

for augment matrix (A)

Similar as Q. 14

Q.17. Verify cayley-hamilton theorem for the matrix

V.V.I

$$\begin{bmatrix} 0 & c & b \\ -c & 0 & a \\ b & -a & 0 \end{bmatrix}$$

=> Given;

let
$$A = \begin{bmatrix} 0 & c & b \\ -c & 0 & a \\ b & -a & 0 \end{bmatrix}$$

The characteristic equation of matrix is -

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 0-\lambda & c & b \\ -c & 0-\lambda & a \\ b & -a & 0-\lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} -\lambda & c & b \\ -c & -\lambda & a \\ b & -a & -\lambda \end{vmatrix} = 0$$

$$-\lambda(\lambda^2 + a^2) - c[c\lambda - ab] + b[ac + b\lambda] = 0$$

$$-\lambda^3 - \lambda a^2 - c^2\lambda + abc + abc + b^2\lambda = 0$$

$$-\lambda^3 - \lambda(a^2 + c^2 - b^2) + 2abc = 0$$

$$\lambda^3 + \lambda(a^2 + c^2 - b^2) - 2abc = 0$$

To verify Cayley hamilton theorem, we have;

$$A^3 + A(a^2+c^2-b^2) - 2abcI = 0$$

Now,

$$A^2 = \begin{bmatrix} 0 & c & b \\ -c & 0 & a \\ b & -a & 0 \end{bmatrix} \begin{bmatrix} 0 & c & b \\ -c & 0 & a \\ b & -a & 0 \end{bmatrix}$$

$$= \begin{bmatrix} b^2-c^2 & -ab & ac \\ ab & -c^2-a^2 & -bc \\ ac & bc & b^2-a^2 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} b^2-c^2 & -ab & ac \\ ab & -c^2-a^2 & -bc \\ ac & bc & b^2-a^2 \end{bmatrix} \begin{bmatrix} 0 & c & b \\ -c & 0 & a \\ b & -a & 0 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} abc+abc & b^2c-c^3-a^2c & b^3-bc^2-a^2b \\ c^3+a^2c-b^2c & abc+abc & ab^2-ac^2-a^3 \\ -bc^2+b^3-a^2b & ac^2+a^3-ab^2 & abc+abc \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 2abc & b^2c-c^3-a^2c & b^3-bc^2-a^2b \\ c^3-b^2c+a^2c & 2abc & ab^2-a^3-ac^2 \\ b^3-bc^2-a^2b & a^3+ac^2-ab^2 & 2abc \end{bmatrix}$$

$$A(a^2+c^2-b^2) = (a^2+c^2-b^2)\begin{bmatrix} 0 & c & b \\ -c & 0 & a \\ b & -a & 0 \end{bmatrix}$$

$$\cancel{= \begin{bmatrix} 0 & a^2c+c^3-b^2c & bc+a^2b-b^3 \\ a^2c & & \end{bmatrix}}$$

$$= \begin{bmatrix} 0 & a^2c+c^3-b^2c & ba^2+bc^2-b^3 \\ +b^2c-c^3-ca^2 & 0 & a^3+ac^2-ab^2 \\ a^2b+c^2b-b^3 & +ab^2 - a^3-ac^2 & 0 \end{bmatrix}$$

$$2abc\, I = \begin{bmatrix} 2abc & 0 & 0 \\ 0 & 2abc & 0 \\ 0 & 0 & 2abc \end{bmatrix}$$

$$A^3 + A(a^2+c^2-b^2) - 2abc\, I = 0$$

$$\begin{bmatrix} 2abc & b^2c-c^3-a^2c & b^3-bc^2-a^2b \\ c^3-b^2c+a^2c & 2abc & ab^2-a^3-ac^2 \\ b^3-bc^2-ba^2 & a^3+ac^2-ab^2 & 2abc \end{bmatrix} +$$

$$\begin{bmatrix} 0 & a^2c+c^3-b^2c & ba^2+bc^2-b^3 \\ b^2c-c^3-ca^2 & 0 & a^3+ac^2-ab^2 \\ a^2b+c^2b-b^3 & ab^2-a^3-ac^2 & 0 \end{bmatrix} - \begin{bmatrix} 2abc & 0 & 0 \\ 0 & 2abc & 0 \\ 0 & 0 & 2abc \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Hence, verified cayley-hamilton theorem.

Q.18 Find the eigen values and eigen vector of the matrix

V.V.I

$$\begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$$

=> Given;

let $A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$

The characteristic equation of given matrix is

$|A - \lambda I| = 0$

$$\begin{vmatrix} 1-\lambda & 1 & 3 \\ 1 & 5-\lambda & 1 \\ 3 & 1 & 1-\lambda \end{vmatrix} = 0$$

$(1-\lambda)[(5-\lambda)(1-\lambda)-1] - 1[1-\lambda-3] +$

$3[1-3(5-\lambda)] = 0$

$(1-\lambda)[5-\lambda-5\lambda+\lambda^2-1] - 1[-\lambda-2] +$

$3[1-15+3\lambda] = 0$

$(1-\lambda)[\lambda^2-6\lambda+4] + [\lambda+2] + 3[3\lambda-14] = 0$

$$\lambda^2 - 6\lambda + 4 - \lambda^3 + 6\lambda^2 - 4\lambda + \lambda + 2 + 9\lambda - 42 = 0$$

$$-\lambda^3 + 7\lambda^2 - 36 = 0$$

$$\lambda^3 - 7\lambda^2 + 36 = 0$$

at $\lambda = -2$ it satisfy the above equation

$\therefore$ $\lambda + 2$ is factor of above equation

$$\begin{array}{r|l} & \lambda^2 - 9\lambda + 18 \\ \hline \lambda + 2 & \lambda^3 - 7\lambda^2 + 36 \\ & -\lambda^3 + 2\lambda^2 \\ & - \quad - \\ \hline & -9\lambda^2 + 36 \\ & -9\lambda^2 - 18\lambda \\ & + \quad + \\ \hline & 18\lambda + 36 \\ & 18\lambda + 36 \\ & - \quad - \\ \hline & \times\times \end{array}$$

$$(\lambda + 2)(\lambda^2 - 9\lambda + 18) = 0$$

$$(\lambda + 2)(\lambda^2 - 6\lambda - 3\lambda + 18) = 0$$

$$(\lambda + 2)(\lambda(\lambda - 6) - 3(\lambda - 6)) = 0$$

$$(\lambda + 2)(\lambda - 6)(\lambda - 3) = 0$$

$\therefore$ the value of λ are $-2, 3, 6$

Hence, these are eigen values of given matrix.

Here, To find eigen vector

at $\lambda = -2$

$$[A - \lambda I]X = 0$$

$$\begin{bmatrix} 1-\lambda & 1 & 3 \\ 1 & 5-\lambda & 1 \\ 3 & 1 & 1-\lambda \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 1 & 3 \\ 1 & 7 & 1 \\ 3 & 1 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$3x + y + 3z = 0$ —— (i)
$x + 7y + z = 0$ —— (ii)
$3x + y + 3z = 0$ —— (iii)

Here, first and third equation is same, we have first two equation;

$3x + y + 3z = 0$
$x + 7y + z = 0$

$$\frac{x}{\begin{vmatrix} 1 & 3 \\ 7 & 1 \end{vmatrix}} = \frac{-y}{\begin{vmatrix} 3 & 3 \\ 1 & 1 \end{vmatrix}} = \frac{z}{\begin{vmatrix} 3 & 1 \\ 1 & 7 \end{vmatrix}} = k_1 \text{ (let)}$$

$$\frac{x}{-20} = \frac{-y}{0} = \frac{z}{20} = k_1$$

$$\frac{x}{-1} = \frac{y}{0} = \frac{z}{1} = k_1$$

$k_1(-1, 0, 1)$ is the eigen vector at $\lambda = -2$.

eigent vector;

at $\lambda = 3$

$$\begin{bmatrix} -2 & 1 & 3 \\ 1 & 2 & 1 \\ 3 & 1 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$-2x + y + 3z = 0$ —— (iv)
$x + 2y + z = 0$ —— (v)
$3x + y - 2z = 0$ —— (vi)

from eq- (iv) and (v);

$$\frac{x}{\begin{vmatrix} 1 & 3 \\ 2 & 1 \end{vmatrix}} = \frac{-y}{\begin{vmatrix} -2 & 3 \\ 1 & 1 \end{vmatrix}} = \frac{z}{\begin{vmatrix} -2 & 1 \\ 1 & 2 \end{vmatrix}} = k_2 \text{ (let)}$$

$$\frac{x}{-5} = \frac{-y}{-5} = \frac{z}{-5} = k_2$$

$$\frac{x}{-1} = \frac{y}{1} = \frac{z}{-1} = k_2$$

$k_2(-1, 1, 1)$ is the eigen vector at $\lambda = 3$

at $\lambda = 6$

$$\begin{bmatrix} -5 & 1 & 3 \\ 1 & -1 & 1 \\ 3 & 1 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$-5x + y + 3z = 0$ —— (vii)

$x - y + z = 0$ —— (viii)

$3x + y - 5z = 0$ —— (ix)

from eq- (vii) and (viii)

$-5x + y + 3z = 0$

$x - y + z = 0$

$$\frac{x}{\begin{vmatrix} 1 & 3 \\ -1 & 1 \end{vmatrix}} = \frac{-y}{\begin{vmatrix} -5 & 3 \\ 1 & 1 \end{vmatrix}} = \frac{z}{\begin{vmatrix} -5 & 1 \\ 1 & -1 \end{vmatrix}} = k_3 \text{ (say)}$$

$$\frac{x}{4} = \frac{-y}{-8} = \frac{z}{4}$$

$$\frac{x}{1} = \frac{y}{2} = \frac{z}{1} = k_3$$

$k_3(1, 2, 1)$ is the eigen vector ~~of~~ at $\lambda = 6$.

Hence eigen values are $-2, 3, 6$
and
eigen vectors are $(-1, 0, 1), (-1, 1, -1), (1, 2, 1)$

Q.19. Find the eigen values and eigen vectors of the following matrix :

VVI.

$$\begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$$

$\Rightarrow$ Given;

let $A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$

The characteristic equation of given matrix is

$|A - \lambda I| = 0$

$$\begin{vmatrix} 6-\lambda & -2 & 2 \\ -2 & 3-\lambda & -1 \\ 2 & -1 & 3-\lambda \end{vmatrix} = 0$$

$(6-\lambda)[(3-\lambda)(3-\lambda) - 1] + 2[-2(3-\lambda) + 2]$
$+ 2[2 - 2(3-\lambda)] = 0$

$(6-\lambda)[9 + \lambda^2 - 6\lambda - 1] + 2[2\lambda - 6 + 2] +$
$2[2 - 6 + 2\lambda] = 0$

$$(6-\lambda)[\lambda^2-6\lambda+8]+2(2\lambda-4)+2(2\lambda-4)=0$$

$$(6-\lambda)[\lambda^2-4\lambda-2\lambda+8]+4(2\lambda-4)=0$$

$$(6-\lambda)[\lambda(\lambda-4)-2(\lambda-4)]+8(\lambda-2)=0$$

$$(6-\lambda)[\lambda-4)[\lambda-2]+8(\lambda-2)=0$$

$$(\lambda-2)[[6-\lambda][\lambda-4]+8]=0$$

$$(\lambda-2)[6\lambda-\lambda^2-24+4\lambda+8]=0$$

$$(\lambda-2)[-\lambda^2+10\lambda-16]=0$$

$$(\lambda-2)(\lambda^2-10\lambda+16]=0$$

$$[\lambda-2][\lambda^2-8\lambda-2\lambda+16]=0$$

$$[\lambda-2][\lambda(\lambda-8)-2(\lambda-8)]=0$$

$$[\lambda-2][\lambda-8][\lambda-2]=0$$

$$\underline{\lambda=2,2,8}$$

Hence, these are eigen values of given matrix

To find eigen vector ;

~~at $\lambda = 8$~~

$$[A - \lambda I]X = 0$$

$$\begin{bmatrix} 6-\lambda & -2 & 2 \\ -2 & 3-\lambda & -1 \\ 2 & -1 & 3-\lambda \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

at $\lambda = 8$

$$\begin{bmatrix} -2 & -2 & 2 \\ -2 & -5 & -1 \\ 2 & -1 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$-2x - 2y + 2z = 0$ —— (i)
$-2x - 5y - z = 0$ —— (ii)
$2x - y - 5z = 0$ —— (iii)

from eq - (i) and (ii)

$-2x - 2y + 2z = 0$
$-2x - 5y - z = 0$

$$\frac{x}{\begin{vmatrix} -2 & 2 \\ -5 & -1 \end{vmatrix}} = \frac{-y}{\begin{vmatrix} -2 & 2 \\ -2 & -1 \end{vmatrix}} = \frac{z}{\begin{vmatrix} -2 & -2 \\ -2 & -5 \end{vmatrix}} = k_1 \text{ (let)}$$

$$\frac{x}{12} = \frac{-y}{6} = \frac{z}{6} = k_1 \text{ (let)}$$

$$\frac{x}{2} = \frac{y}{-1} = \frac{z}{1} = k_1$$

$k_1(2,-1,1)$ is the eigen vector at $\lambda = 8$.

at $\lambda = 2$

$$\begin{bmatrix} 4 & -2 & 2 \\ -2 & 1 & -1 \\ 2 & -1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$4x - 2y + 2z = 0$ —— (iv)

$-2x + y - z = 0$ —— (v)

$2x - y + z = 0$ —— (vi)

all these three equations are same, so we take some parameters

let $x = P_1$

$z = P_2$

~~from eq (i)~~

~~$4x - 2P_1 + 2P_2 = 0$~~

~~$2x - P_1 + P_2 = 0$~~

~~$y = \frac{P_1 + P_2}{2}$~~

from eq- (iv)

$$4P_1 - 2y + 2P_2 = 0$$

$$2P_1 - y + P_2 = 0$$

$$y = 2P_1 + P_2$$

Here,

$$x = P_1$$
$$y = 2P_1 + P_2$$
$$z = P_2$$

$$\therefore \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} P_1 \\ 2P_1 + P_2 \\ P_2 \end{bmatrix}$$

$$= P_1 \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} + P_2 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

Hence eigen vector of $\lambda = 2$ are

$$\begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} \& \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

Hence, the eigen values are 2, 2, 8

the eigen vector are (1, 2, 0), (0, 1, 1), (2, -1, 1).

Q.20 Determine the eigen values and eigen vectors of the following matrix:

V.V.I

$$\begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$$

$\Rightarrow$ Given;

$$A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$$

The characteristic equation of given matrix is

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} -2-\lambda & 2 & -3 \\ 2 & 1-\lambda & -6 \\ -1 & -2 & 0-\lambda \end{vmatrix} = 0$$

$$(-2-\lambda)\left[(1-\lambda)(-\lambda)-12\right] - 2\left[-2\lambda - 6\right]$$
$$-3\left[-4+(1-\lambda)\right] = 0$$

$$(-2-\lambda)\left[-\lambda+\lambda^2-12\right] + 2\left(2\lambda+6\right]$$
$$-3(-3-\lambda) = 0$$

$$(-2-\lambda)[\lambda^2-\lambda-12] + 4(\lambda+3) + 3(\lambda+3) = 0$$

$$(-2-\lambda)[\lambda^2-4\lambda+3\lambda-12] + 7(\lambda+3) = 0$$

$$(-2-\lambda)(\lambda(\lambda-4)+3(\lambda-4)) + 7(\lambda+3) = 0$$

$$(-2-\lambda)(\lambda-4)(\lambda+3) + 7(\lambda+3) = 0$$

$$(\lambda+3)((-2-\lambda)(\lambda-4)+7) = 0$$

$$(\lambda+3)[-2\lambda-\lambda^2+8+4\lambda+7] = 0$$

$$[\lambda+3][-\lambda^2+2\lambda+15] = 0$$

$$[\lambda+3][\lambda^2-2\lambda-15] = 0$$

$$[\lambda+3][\lambda^2+3\lambda-5\lambda-15] = 0$$

$$[\lambda+3][\lambda(\lambda+3)-5(\lambda+3)] = 0$$

$$(\lambda+3)[\lambda+3][\lambda-5] = 0$$

$$\lambda = -3, -3, 5$$

Hence eigen values are $-3, -3, 5$.

To find eigen vector;

$$[A-\lambda I]X = 0$$

$$\begin{bmatrix} -2-\lambda & 2 & -3 \\ 2 & 1-\lambda & -6 \\ -1 & -2 & 0-\lambda \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

at $\lambda = 5$

$$\begin{bmatrix} -7 & 2 & -3 \\ 2 & -4 & -6 \\ -1 & -2 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$-7x + 2y - 3z = 0$ —— (i)
$2x - 4y - 6z = 0$ —— (ii)
$-x - 2y - 5z = 0$ —— (iii)

from eq- (ii) and (iii),

$x - 2y - 3z = 0$
$-x - 2y - 5z = 0$

$$\frac{x}{\begin{vmatrix} -2 & -3 \\ -2 & -5 \end{vmatrix}} = \frac{-y}{\begin{vmatrix} 1 & -3 \\ -1 & -5 \end{vmatrix}} = \frac{z}{\begin{vmatrix} 1 & -2 \\ -1 & -2 \end{vmatrix}} = k_1 \text{ (let)}$$

$$\frac{x}{4} = \frac{-y}{-8} = \frac{z}{-4} = k_1$$

$$\frac{x}{1} = \frac{y}{2} = \frac{z}{-1} = k_1$$

Hence $k_1(1, 2, -1)$ is the eigen vector at $\lambda = 5$

at $\lambda = -3$

$$\begin{bmatrix} 1 & 2 & -3 \\ 2 & 4 & -6 \\ -1 & -2 & +3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$x + 2y - 3z = 0$ —— (iv)
$2x + 4y - 6z = 0$ —— (v)
$-x - 2y + 3z = 0$ —— (vi)

all these three equation are same
$\therefore$ we take some parameters

let $y = P_1$
$z = P_2$

from eq – (iv)

$x + 2y - 3z = 0$
$x + 2P_1 - 3P_2 = 0$
$x = 3P_2 - 2P_1$

Here,

$$x = 3P_2 - 2P_1$$
$$y = P_1$$
$$z = P_2$$

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3P_2 - 2P_1 \\ P_1 \\ P_2 \end{bmatrix}$$

$$= P_1 \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + P_2 \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}$$

Hence the eigen vector

at $\lambda = -3$ are $(-2,1,0)$ & $(3,0,1)$

Hence, eigen values are $-3, -3, 5$

eigen vectors are $(-2,1,0), (3,0,1), (1,2,-1)$

{Note:- If same eigen vector then diagonalization not possible.}

Q.21. Examine whether the matrix

V.V.I

$$A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$$ is

diagonalizable. If so, obtain the matrix P such that $P^{-1}AP$ is a diagonal matrix.

⇒ Given;

$$A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$$

The characteristic equation of matrix is

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} -2-\lambda & 2 & -3 \\ 2 & 1-\lambda & -6 \\ -1 & -2 & 0 \end{vmatrix}$$

This is same as Q. 20.//

So eigen values are; $-3, -3, 5$

eigen vector are $(-2,1,0), (3,0,1)\ (1,2,-1)$

Eigen vector $\begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$

writting the three eigen vector as the three column, we get modal matrix (P);

$$P = \begin{bmatrix} -2 & 3 & 1 \\ 1 & 0 & 2 \\ 0 & 1 & -1 \end{bmatrix}$$

and

diagonal matrix are; ($\lambda_1, \lambda_2, \lambda_3 \rightarrow$ eigen values)

$$D = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix}$$

$$D = \begin{bmatrix} -3 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

Q22. Reduce the matrix $A = \begin{bmatrix} -1 & 2 & -2 \\ 1 & 2 & 1 \\ -1 & -1 & 0 \end{bmatrix}$

V.V.I

to the diagonal form.

$\Rightarrow$ Given

$$A = \begin{bmatrix} -1 & 2 & -2 \\ 1 & 2 & 1 \\ -1 & -1 & 0 \end{bmatrix}$$

The characteristic equation of given matrix is

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} -1-\lambda & 2 & -2 \\ 1 & 2-\lambda & 1 \\ -1 & -1 & 0-\lambda \end{vmatrix} = 0$$

$$(-1-\lambda)[(2-\lambda)(-\lambda)+1] - 2[-\lambda+1] - 2[-1+2-\lambda] = 0$$

$$(-1-\lambda)[-2\lambda+\lambda^2+1] - 2[1-\lambda] - 2[1-\lambda] = 0$$

$$(-1-\lambda)[\lambda^2-\lambda-\lambda+1] - 4[1-\lambda] = 0$$

$$(-1-\lambda)[\lambda(\lambda-1)-1(\lambda-1)] + 4(\lambda-1) = 0$$

$$(-1-\lambda)[\lambda-1][\lambda-1] + 4[\lambda-1] = 0$$

$$[\lambda-1)[(-1-\lambda)(\lambda-1)+4] = 0$$

$$(\lambda-1)[-\lambda-\lambda^2+1+\lambda+4] = 0$$

$(\lambda - 1)[5 - \lambda^2] = 0$

$\lambda = 1$ or $\lambda^2 = 5$

$\lambda = \pm\sqrt{5}$

$\lambda = 1, -\sqrt{5}, \sqrt{5}$

Hence this is the eigen value of given matrix.

to find eigen vector ;

$[A - \lambda I] X = 0$

$$\begin{bmatrix} -1-\lambda & 2 & -2 \\ 1 & 2-\lambda & 1 \\ -1 & -1 & 0-\lambda \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

at $\lambda = 1$

$$\begin{bmatrix} -2 & 2 & -2 \\ 1 & 1 & 1 \\ -1 & -1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$-2x + 2y - 2z = 0$ —— (i)

$x + y + z = 0$ —— (ii)

$-x - y - z = 0$ —— (iii)

~~for~~ Second and third equation are same
$\therefore$ we take first two equation

$$-2x + 2y - 2z = 0$$
$$x + y + z = 0$$

$$\frac{x}{\begin{vmatrix} 2 & -2 \\ 1 & 1 \end{vmatrix}} = \frac{-y}{\begin{vmatrix} -2 & -2 \\ 1 & 1 \end{vmatrix}} = \frac{z}{\begin{vmatrix} -2 & 2 \\ 1 & 1 \end{vmatrix}} = k_1 \text{ (let)}$$

$$\frac{x}{4} = \frac{-y}{0} = \frac{z}{-4} = k_1$$

$$\frac{x}{1} = \frac{y}{0} = \frac{z}{1} = k_1$$

vector

the eigen ~~value~~ $k_1(1, 0, 1)$ at $\lambda = 1$

at $\lambda = -\sqrt{5}$

$$\begin{bmatrix} \sqrt{5}-1 & 2 & -2 \\ 1 & 2+\sqrt{5} & 1 \\ -1 & -1 & \sqrt{5} \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$(\sqrt{5}-1)x + 2y - 2z = 0 \quad \text{— (iv)}$$
$$x + (2+\sqrt{5})y + z = 0 \quad \text{— (v)}$$
$$-x - y + \sqrt{5}z = 0 \quad \text{— (vi)}$$

from eq- (v) and (vi)

$$x + (2+\sqrt{5})y + z = 0$$
$$-x - y + \sqrt{5}z = 0$$

$$\frac{x}{\begin{vmatrix} 2+\sqrt{5} & 1 \\ -1 & \sqrt{5} \end{vmatrix}} = \frac{-y}{\begin{vmatrix} 1 & 1 \\ -1 & \sqrt{5} \end{vmatrix}} = \frac{z}{\begin{vmatrix} 1 & 2+\sqrt{5} \\ -1 & -1 \end{vmatrix}} = k_2 \text{ (say)}$$

$$\frac{x}{2\sqrt{5}+5+1} = \frac{-y}{\sqrt{5}+1} = \frac{z}{-1+2+\sqrt{5}} = k_2$$

$$\cancel{\frac{x}{2\sqrt{5}+6} = \frac{-y}{\sqrt{5}+1} = \frac{z}{\sqrt{5}+1}}$$

$$\frac{x}{1^2+(\sqrt{5})^2+2\sqrt{5}} = \frac{-y}{\sqrt{5}+1} = \frac{z}{\sqrt{5}+1} = k_2$$

$$\frac{x}{(\sqrt{5}+1)^2} = \frac{-y}{\sqrt{5}+1} = \frac{z}{\sqrt{5}+1} = k_2$$

$$\frac{x}{\sqrt{5}+1} = \frac{y}{-1} = \frac{z}{1} = k_2$$

$\therefore$ The eigen vector $k_2(\sqrt{5}+1, -1, 1)$ at

$\lambda = -\sqrt{5}$

Similarly

at $\lambda = \sqrt{5}$

the eigen vector are $(\sqrt{5}-1, 1, -1)$.

Now, writing the three eigen vectors as the three column, we obtain modal matrix (P)

$$P = \begin{bmatrix} 1 & \sqrt{5}+1 & \sqrt{5}-1 \\ 0 & -1 & 1 \\ 1 & 1 & -1 \end{bmatrix}$$

and

Diagonal matrix is

$$D = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix}$$

$$D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -\sqrt{5} & 0 \\ 0 & 0 & +\sqrt{5} \end{bmatrix}$$